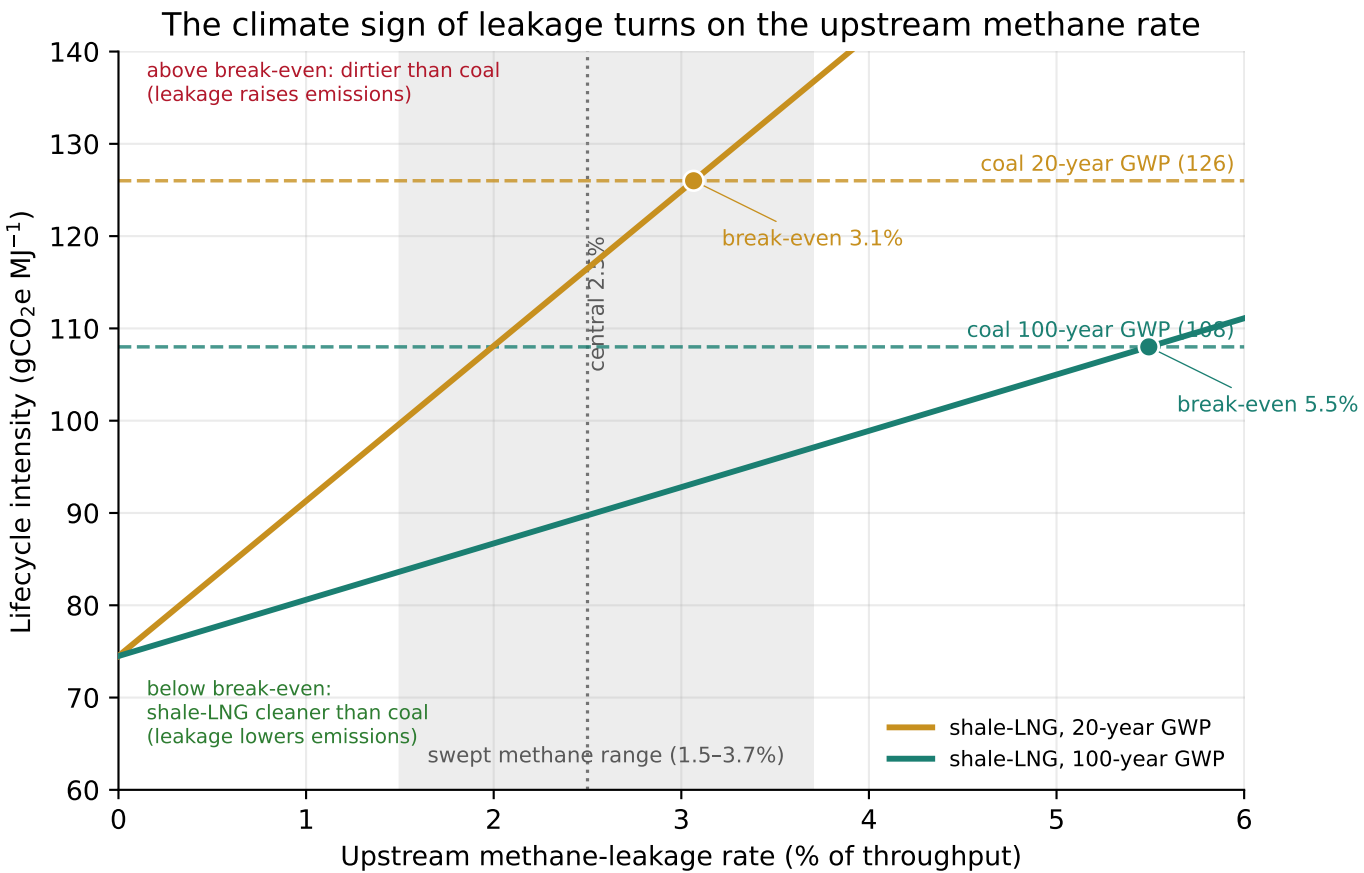
The climate sign of leakage turns on the upstream methane rate
above break-even: dirtier than coal
(leakage raises emissions)
coal 20-year GWP (126)
break-even 3.1%
central 2.5%
coal 100-year GWP (108)
break-even 5.5%
below break-even:
shale-LNG cleaner than coal
(leakage lowers emissions)
swept methane range (1.5–3.7%)
shale-LNG, 20-year GWP
shale-LNG, 100-year GWP
Lifecycle intensity (gCO2e MJ−1)
Upstream methane-leakage rate (% of throughput)
140
130
120
110
100
90
80
70
60
0
1
2
3
4
5
6